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Representative Learning for Distributed Learning with Heterogeneity and Asynchrony

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ABSTRACT

Representative Learning (RepL) is a distributed learning framework in which nodes transmit pseudo data, called representatives, instead of model parameters or gradients. These representatives retain the original data format while encoding key statistical features, enabling them to support asynchronous communication and heterogeneous tasks. This paper introduces two new representative constructions: the Transformed Mean Representative (TMR), which generalizes the mean representative by incorporating model-specific link functions; and the Anchored Score-Matching Representative (Anchored-SMR), which modifies score-matching equations to ensure uniqueness and stability. Anchored-SMR is further extended to accommodate general smooth loss functions with optional non-smooth penalties. We analyze RepL in decentralized, asynchronous systems where gradients and models from other nodes may be delayed or misaligned. Theoretical results and extensive simulations demonstrate that the proposed representatives maintain accuracy and convergence under heterogeneity and asynchrony, offering a scalable and interpretable alternative to gradient-based distributed optimization.

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Decentralized learning;
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Subpartitioning;
Transformed mean representative

1. Introduction

The rise of Artificial Intelligence and Machine Learning has fueled demand for Distributed Machine Learning (DML), which enables collaborative training across decentralized nodes while preserving data privacy and reducing communication costs. DML is crucial in applications such as healthcare, finance, and autonomous systems, where centralized data collection is infeasible or undesirable due to privacy, bandwidth, or legal constraints.

Foundational DML frameworks include MapReduce (Dean and Ghemawat 2008), parallel stochastic gradient descent (SGD) (Zinkevich et al. 2010), and the Parameter Server architecture (Li et al. 2014), all of which laid the groundwork for scalable optimization in distributed settings.

DML faces fundamental challenges: data heterogeneity, communication bottlenecks, privacy constraints, and instability under delayed or missing updates. Data across nodes is often non-IID, with substantial variability in feature or label distributions, impeding model generalization. Communication costs escalate with frequent updates, especially under limited bandwidth. Privacy regulations (e.g., GDPR) restrict data transfer, and asynchronous updates introduce convergence issues under model drift or node failures.

Federated Learning (FL) (Konečný et al. 2016) addresses many DML concerns by training models locally and sharing only parameter updates. Enhancements such as model compression and sparse updates reduce communication (Stich 2018; Stich and Karimireddy 2020; Spiridonoff, Olshevsky, and Paschalidis 2021; Wang and Joshi 2021). However, FL is vulnerable to performance degradation under heterogeneity (Zhou and Cong 2018; Haddadpour and Mahdavi 2019; Khaled, Mishchenko, and Richtárik 2020; Koloskova et al. 2020; Sery et al. 2021; Wang et al. 2021; Zhu, Hong, and Zhou 2021; Fang and Ye 2022; Liu et al. 2022; Mendieta et al. 2022; Ye et al. 2024).

To mitigate this, personalized FL (Fallah, Mokhtari, and Ozdaglar 2020; Li et al. 2021; Tan et al. 2023) and multi-task learning (Argyriou, Evgeniou, and Pontil 2006; Zhang and Yeung 2012; Smith et al. 2017; Corinzia, Beuret, and Buhmann 2019) aim to tailor models per node. FL has expanded to hierarchical and peer-to-peer architectures (Lian et al. 2017; Assran et al. 2019) and been applied in healthcare, mobile systems, and finance (Zhao et al. 2018; Li et al. 2020; Cheng et al. 2021).

Recent work addresses heterogeneity and asynchrony through contrastive objectives, robustness mechanisms, and representation alignment. For example, relaxed contrastive learning such as FedRCL (Seo et al. 2024) improves cross-client feature alignment under distribution shift via latent consistency. Robust FL methods (Fang, Ye, and Du 2025) mitigate corrupted or unreliable clients through asymmetric aggregation or outlier-resistant updates, while CLIP-guided federated learning (Shi et al. 2024) leverages pretrained multimodal representations to stabilize training under severe mismatch. Although these methods enhance robustness or representation quality, they still rely on model- or gradient-based communication and generally assume synchronized rounds, task-aligned semantics, or fixed architectures.

Model-agnostic approaches, such as knowledge distillation (Chang et al. 2019; Lin et al. 2020; Wu et al. 2022), allow nodes to transmit soft predictions, though they often assume task-aligned semantics and synchronized updates. Distributed statistical inference (Zhang, Duchi, and Wainwright 2013; Battey et al. 2018; Fan, Guo, and Wang 2023) enables one-shot aggregation

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via sufficient statistics or estimating equations, but is typically limited to GLMs or linear models with aligned objectives.

Most DML methods rely on *model-derived summaries*, such as gradients, logits, or sufficient statistics, as communication units. However, these are tightly entangled with the local model architecture, task formulation, and update timing. When assumptions of task alignment, model compatibility, or synchronization fail, as in heterogeneous or asynchronous systems, these summaries become misleading or unusable, limiting their relevance and reusability.

Moreover, such summaries are not traceable or interpretable in raw data terms and often discard structural context upon aggregation. This motivates a shift from transmitting insights to communicating structured *knowledge*, pseudo data that preserves original format and local statistical properties.

Representative Learning (RepL) adopts this principle by constructing *representatives* at each node: pseudo samples that encode key features of local data in its native form. These structured summaries are model-agnostic, support asynchronous and decentralized settings, and enable transparency in collaborative learning.

To illustrate the necessity of such an approach, consider a binary classification setting with severe *label support mismatch*, where some nodes observe almost exclusively one class (e.g., $P_k(Y = 1) \approx 0$) while others observe the opposite, even though the conditional relationship $P(Y \mid X)$ is shared globally. In this case, local logistic regression models are ill-conditioned or degenerate, and gradient-based federated aggregation amplifies instability rather than correcting it. RepL circumvents this failure mode by transmitting representative pseudo-data that preserve conditional structure across nodes, allowing global learning to proceed even when local model fitting is unreliable.

In the remainder of this paper, Section 2 introduces the RepL framework. Section 3 develops representative constructions under GLMs. Section 4 generalizes RepL to smooth and non-smooth losses with optional regularization. Section 5 evaluates robustness under decentralization and asynchrony. Section 6 presents simulation results, Section 7 gives a real-world case study, and Section 8 concludes with future directions.

2. Representative Learning Architecture

RepL replaces model-derived summaries (e.g., weights, gradients) with structured pseudo-data, or *representatives*, that match statistical properties of local datasets while preserving the original format. This shift enables communication without model forms, objectives, or synchronization, and supports collaboration in heterogeneous or decentralized systems.

2.1. Formulation of Representative Learning

Consider a distributed dataset $D = \{D_k\}_{k=1}^K$, where each local dataset $D_k = \{(y_i, \mathbf{x}_i), \mathbf{x}_i \in \mathbb{R}^p, i \in I_k\}$ contains $n_k = I_k$ samples on node k. The global objective seeks to minimize

$$\min_{\boldsymbol{\beta}} f(\boldsymbol{\beta}; D) = \sum_{k=1}^{K} l(\boldsymbol{\beta}; D_k) + r(\boldsymbol{\beta}; \lambda), \tag{1}$$

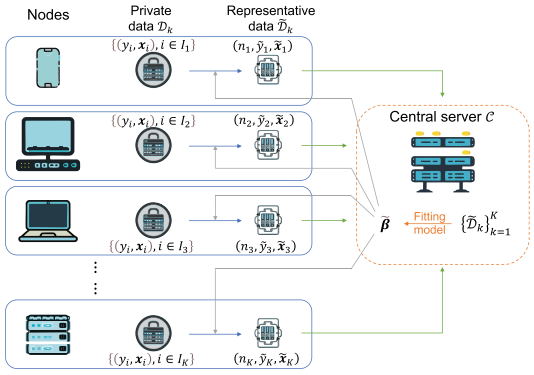


Figure 1. Architecture of RepL.

across all distributed nodes either through a central server $\mathcal{C}$ or decentralized coordination among nodes. Here, $l(\boldsymbol{\beta}; D_k)$ is the local loss or log-likelihood function and $r(\boldsymbol{\beta}; \lambda)$ is a regularization term, often used to penalize the discrepancy between local models or to enforce global model regularity.

The local objective at node k is defined as $f_k(\boldsymbol{\beta}_k; D_k) = l(\boldsymbol{\beta}_k; \mathbf{y}_k, \mathbf{X}_k) + r(\boldsymbol{\beta}_k; \lambda)$. Traditional federated optimization solves this by alternating between local updates and centralized aggregation, which becomes fragile under data heterogeneity or asynchronous schedules.

RepL constructs representatives $\tilde{D}_k = (n_k, \tilde{y}_k, \tilde{\mathbf{x}}_k)$, designed to approximate the statistical structure (e.g., moment or gradient conditions) of D_k under the same data format. These representatives are then used in place of raw data or model weights to compute global updates, for example,

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \eta \sum_{k=1}^{K} \frac{n_k}{n} \nabla l(\boldsymbol{\beta}^{(t)}; \tilde{D}_k).$$

This design decouples distributed optimization from model structure, objective alignment, and synchronization, enabling learning in heterogeneous, asynchronous, or partially connected systems. See Figure 1 for an illustration.

2.2. Advantages of Representative Learning

RepL replaces gradient or weight transmission with pseudo-data representatives that retain the original data format, enabling compatibility across heterogeneous tasks and models. Their construction allows asynchronous communication without global coordination, as the updates are not tied to transient model states. Communication is efficient, with each representative consisting of a single compact tuple per node. Moreover, representatives can be stored and reused across rounds, supporting memory-based learning, interpretability, and resilience to partial participation.

2.3. Advancements in Representative Learning

RepL methods differ in how they approximate local objective structure. The Mean Representative (MR) summarizes each node by its sample means, offering a model-free baseline suitable for linear objectives. The Score-Matching Representative (SMR)

extends this to Generalized Linear Models (GLMs) by solving moment equations that match local score (gradient) functions. In GLMs, the maximum likelihood estimate solves the score equation,

$$s(\boldsymbol{\beta}; D) = \sum_{i=1}^{N} (y_i - G(\eta_i))\nu(\eta_i)\mathbf{x}_i = 0, \tag{2}$$

where the linear predictor is $\eta_i = \mathbf{x}_i^T\boldsymbol{\beta}$, and $G(\cdot)$ is the inverse function of the link function $g(\cdot)$. The term residual $r_i = y_i - G(\eta_i)$ represents the excess response over the expectation, and its product with $\nu(\eta) = a(\phi)^{-1}\theta'(\eta)$ measures the contribution of the observation. SMR approximates this gradient structure via pseudo-data, achieving $O(\Delta^{1/2})$ convergence when covariates are well-clustered (Li and Yang 2022). Response-Aided SMR (RASMR) improves robustness under heterogeneity by incorporating both η_i and residuals r_i into subpartitioning, aligning representatives with local gradient structure (Zheng, Li, and Yang 2024).

Recent RepL extensions incorporate residual-guided subpartitioning and model-specific structure to improve fidelity under heterogeneity. The DCL framework (Casey et al. 2025) uses representatives to transfer calibrated knowledge between mismatched tasks, weighting transferred predictions by alignment scores. Deep RepL (Fan et al. 2024) uses representative batches to stabilize training of distributed neural networks. These innovations support RepL's adaptability across statistical and deep models.

Supplementary Section S.1 provides complete GLM expressions and the representative matching formulations used in these frameworks.

3. Transformed and Anchored Representatives

As data heterogeneity increases, classical representatives face limitations: MR fails under nonlinearity, and SMR may yield unstable intercepts and drifted covariates. We propose two improved constructions: the **Transformed Mean Representative (TMR)**, which adapts MR using link-function transformations for better GLM compatibility, and the **Anchored Score-Matching Representative (Anchored-SMR)**, which improves upon SMR by anchoring intercept estimation, enhancing stability and interpretability.

3.1. Transformed Mean Representative (TMR)

MR method is effective for linear models but inadequate for GLMs, where the link function introduces nonlinear response-covariate relationships. In such cases, naive averaging can yield biased estimates, especially under steep inverse link functions $G(\cdot)$, as seen in Poisson regression with a log link.

To address this, we introduce the Transformed Mean Representative (TMR). While the covariate representative $\tilde{x}_{kj}$ remains the arithmetic mean, TMR adjusts the response representative $\tilde{y}_k$ by incorporating the link function $g(\cdot)$,

$$\tilde{y}_k = G(n_k^{-1}\sum_{i\in I_k} g(y_i)), \quad \tilde{x}_{kj} = n_k^{-1}\sum_{i\in I_k} x_{ij}, \quad j = 1, \ldots, p.$$

When $g(y_i)$ is undefined (e.g., binary responses), TMR defaults to MR to ensure stability.

TMR reduces bias where $G(\cdot)$ has high curvature or local variability is large. Supplementary Section S.6 includes worked examples.

3.1.1. Error Bound

Let $\hat{\boldsymbol{\beta}}$ be the MLE from full data, and $\tilde{\boldsymbol{\beta}}_{\text{TMR}}$ be the estimate from TMR representatives. Denote by $\Delta := \max_k \max_{i\in I_k} |\mathbf{x}_i - \tilde{\mathbf{x}}_k|$ the maximum within-node covariate deviation, and $\delta := \max_k |\bar{y}_k - \tilde{y}_k|$ the distortion in response introduced by the TMR transformation.

The link function is essential for controlling response distortion in TMR: averaging on the transformed scale $g(y)$ and mapping back via $G = g^{-1}$ linearizes the nonlinear mean structure of the GLM, enabling the local deviation between original and representative responses to be bounded by the local spread of $g(y_i)$ under mild smoothness assumptions.

Theorem 3.1. Assume the log-likelihood $l(\boldsymbol{\beta})$ for the GLM is strictly concave and twice differentiable. Then, as the maximum within-node covariate deviation $\Delta \to 0$, we have

$$\tilde{\boldsymbol{\beta}}_{\text{TMR}} - \hat{\boldsymbol{\beta}}_2 = O(\Delta^{1/2}).$$

The proof is given in Section S.9 of the Supplementary Material. This result shows that the discrepancy between TMR and full-data estimates vanishes at rate $O((\delta + \Delta)^{1/2})$, where Δ measures the maximum within-node covariate deviation and δ bounds the deviation between original responses and the TMR-constructed response. When the link function g is smooth and locally Lipschitz, averaging on the transformed scale ensures that the response distortion satisfies $\delta = O(\Delta)$, leading to the stated $O(\Delta^{1/2})$ convergence rate. The error reduces as subpartitioning or local homogeneity improves.

3.1.2. Initial Choice: TMR or MR?

TMR extends MR by incorporating transformations from the GLM link function, enabling better approximation in models with nonlinear covariate-response relationships. In cases where the link function is identity, as in Gaussian models, or when all responses within a node are identical, TMR reduces to MR. For binary tasks, however, unbounded g makes TMR degenerate to a hard threshold. As detailed in Supplementary Section S.7.

TMR is particularly effective in models with nonlinear link functions and substantial response variability, where direct averaging induces bias. In such cases, subpartitioning by transformed responses (e.g., $\log(y_i)$ for Poisson), residuals, or linear predictors can enhance local fidelity, yielding representatives that better preserve the structure of local data. TMR is computationally efficient and requires only simple elementwise transformations and averaging, making it a stronger initialization than MR for iterative methods like Anchored-SMR. Simulation results (Section 6.1) confirm that TMR produces more accurate estimates than MR in nonlinear models.

3.2. Anchored Score-Matching Representative (Anchored-SMR)

While SMR provides a model-based approach for constructing representatives, it suffers from several limitations. In GLMs, SMR may produce non-constant intercepts and an underdetermined system of equations, $p+2$ unknowns from $(\tilde{\mathbf{x}}_k, \tilde{y}_k, \tilde{\eta}_k)$ with only $p+1$ constraints from score matching (S.3). To resolve this, Li and Yang (2022) constructed $\tilde{y}_k$ as a local weighted average, but this can introduce bias and restrict solution flexibility. Furthermore, cancelation of residual r_i within nodes may cause covariate representatives to drift away from local geometry, reducing representativeness. In extreme cases, SMR fails to converge due to model misspecification or poor initialization.

To address these challenges, we propose the **Anchored Score-Matching Representative (Anchored-SMR)**, which stabilizes the representative intercept to be constant. Let the linear predictor take the form $\eta_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \beta_0$, with β_0 the intercept. The score-matching equations are redefined as

$$\sum_{i\in I_k} v(\eta_i) r_i \mathbf{x}_i = n_k v(\tilde{\eta}_k) \tilde{r}_k \tilde{\mathbf{x}}_k \tag{3}$$

$$\sum_{i\in I_k} v(\eta_i) r_i = n_k v(\tilde{\eta}_k) \tilde{r}_k. \tag{4}$$

These equations ensure identifiability and yield a unique analytic solution

$$\tilde{\eta}_k = \Big(\sum_{i\in I_k} h(\eta_i, y_i)\Big)^{-1} \sum_{i\in I_k} h(\eta_i, y_i)\eta_i \tag{5}$$

$$\tilde{y}_k = G(\tilde{\eta}_k) + \big(n_k v(\tilde{\eta}_k)\big)^{-1} \sum_{i\in I_k} h(\eta_i, y_i) \tag{6}$$

$$\tilde{\mathbf{x}}_k = \Big(\sum_{i\in I_k} h(\eta_i, y_i)\Big)^{-1} \sum_{i\in I_k} h(\eta_i, y_i)\mathbf{x}_i, \tag{7}$$

where $\tilde{\eta}_k$ and $\tilde{\mathbf{x}}_k$ are both weighted means, with weight $h(\eta_i, y_i) = v(\eta_i) r_i$. This design aligns the representative with node-level geometry and error structure, offering improved accuracy across diverse GLM families (e.g., Bernoulli, Poisson, Gamma, Inverse Gaussian). For worked examples, see Supplementary Section S.8. The Anchored-SMR algorithm is detailed in Algorithm S1.

3.2.1. Update Strategy

When representatives are constructed via MR or TMR, it is appropriate to fully minimize the representative-based log-likelihood, as these summaries encode stationary information near the optimum.

In contrast, Anchored-SMR representatives, constructing from the current model gradient, encode directional information from local score functions, rather than stationary information near the optimum. Fully optimizing Anchored-SMR representatives at each iteration is analogous to using excessively large step sizes in gradient descent. It can overreact to transient gradient signals and hinder convergence. Instead, we adopt a one-step gradient update

$$\tilde{\boldsymbol{\beta}}^{(t+1)} = \tilde{\boldsymbol{\beta}}^{(t)} - \gamma\, s(\tilde{\boldsymbol{\beta}}^{(t)}; \tilde{D}), \tag{8}$$

where γ is a learning rate, and $s(\tilde{\boldsymbol{\beta}}; \tilde{D})$ is the global score computed from Anchored-SMR data. This procedure mirrors SGD under synchronized communication and leverages the directional nature of Anchored-SMR.

The incremental update stabilizes convergence, avoids overreacting to short-term local fluctuations, and reduces both computational and communication costs. As shown in Section 6.3, it accelerates early convergence and remains valid under both centralized and decentralized synchronization. Subpartitioning can further improve fidelity by refining gradient direction locally.

3.2.2. Theoretical Justification of Anchored-SMR

The Anchored-SMR enjoys several desirable theoretical properties. We begin with two propositions that clarify its behavior in special cases.

Proposition 3.1. If $\mathbf{x}_i = \mathbf{x}_1$, for all $i \in I_k$ within node k, then Anchored-SMR coincides with MR,

$$\tilde{\mathbf{x}}_k = \mathbf{x}_1, \quad \tilde{y}_k = \bar{y}_k,$$

where $\bar{y}_k$ is the local average response.

Anchored-SMR constrains the intercept to be constant, maintaining interpretability and ensuring that covariate representatives remain representative, particularly when $h(\eta_i, y_i)$ shares the same sign.

Proposition 3.2. If $h(\eta_i, y_i) = v(\eta_i)(y_i - G(\eta_i))$ shares the same sign for all $i \in I_k$, then the Anchored-SMR representatives satisfy

$$\min_{i\in I_k} x_{ij} \le \tilde{x}_{kj} \le \max_{i\in I_k} x_{ij}, \quad j = 1, \ldots, p, \quad \min_{i\in I_k} \eta_i \le \tilde{\eta}_k \le \max_{i\in I_k} \eta_i.$$

This ensures that representatives remain bounded within local support.

Anchored-SMR preserves key aspects of the original data's log-likelihood landscape. Although the method performs a one-step update per round, it is critical to ensure that the representative data $\tilde{D}$ encodes a comparable statistical structure to the original data D. The following theorem bounds the difference in log-likelihood.

Theorem 3.2. Assume $\theta(\cdot)$, $b(\cdot)$, and $c(\cdot)$ are Lipschitz continuous with constants L_θ, L_b, and L_c, respectively. If within each node the response discrepancy satisfies $|y_i - \tilde{y}_k| \le R_k$ and the within-node covariate deviation is bounded by $\Delta_k := \max_{i\in I_k} \|\mathbf{x}_i - \tilde{\mathbf{x}}_k\|$, then

$$\Big| l(\boldsymbol{\beta}; D) - l(\boldsymbol{\beta}; \tilde{D}) \Big| \le L_1 \sum_{k=1}^{K} n_k R_k + L_2 \|\boldsymbol{\beta}\| \sum_{k=1}^{K} n_k \Delta_k R_k,$$

where the constants $L_1 = \frac{L_\theta}{a(\phi)} + L_c$, and $L_2 = \frac{L_\theta(1+L_b)}{a(\phi)}$.

The proof is provided in Section S.9 of the Supplementary Material. This result shows that the representative dataset $\tilde{D}$ constructed via Anchored-SMR preserves the global likelihood structure of the original data D up to terms involving $\sum_k n_k R_k$ and $\sum_k n_k \Delta_k R_k$. By maintaining relatively uniform residuals within each node and keeping within-node covariate deviations controlled, the discrepancy in log-likelihood remains small. This ensures that the representative dataset captures the statistical

signal of the original data while reducing communication and computation burden.

We now provide theoretical justification for the one-step update algorithm of Anchored-SMR under synchronized communication. This result characterizes its convergence in expectation to a neighborhood of the global minimizer.

Theorem 3.3. Assume the global objective $f(\boldsymbol{\beta}) = \sum_{k=1}^{K} \frac{n_k}{n} l_k(\boldsymbol{\beta})$ is μ-strongly convex and L-smooth. Let each node use representatives $\tilde{D}_k$ such that the surrogate gradient satisfies

$$\|\nabla l_k(\boldsymbol{\beta}) - \nabla \tilde{l}_k(\boldsymbol{\beta})\| \le \varepsilon_k, \quad \text{for all } \boldsymbol{\beta},$$

and define the global representative gradient approximation error $\varepsilon := \sum_k \frac{n_k}{n} \varepsilon_k$. Let the update be

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \eta \sum_{k=1}^{K} \frac{n_k}{n} \nabla \tilde{l}_k(\boldsymbol{\beta}^{(t)}),$$

with step size $\eta \le \min\left\{\frac{\mu}{2L^2}, \frac{1}{4L}\right\}$. Then the iterates satisfy

$$\mathbb{E}\left[\|\boldsymbol{\beta}^{(T)} - \boldsymbol{\beta}^*\|^2\right] \le \left(1 - \tfrac{1}{2}\eta\mu\right)^T \|\boldsymbol{\beta}^{(0)} - \boldsymbol{\beta}^*\|^2 + O\left(\frac{\eta}{\mu}\varepsilon^2\right).$$

The proof is provided in Section S.9 of the Supplementary Material. This result ensures that Anchored-SMR achieves geometric convergence in expectation, up to a residual error governed by ε^2. As local partitions become finer (i.e., $\Delta_k \to 0$), $\varepsilon_k \to 0$, and the updates increasingly approximate those from the full-data objective, preserving statistical efficiency without requiring raw data sharing.

3.2.3. Comparison of TMR, SMR, and Anchored-SMR

TMR, SMR, and Anchored-SMR differ in their construction principles, computational behavior, and ability to handle data heterogeneity. Figure 2 provides a schematic comparison of the representative construction workflows for TMR and Anchored-SMR, highlighting the one-shot versus iterative nature of the two methods.

TMR provides a fast, one-shot representative by matching link-transformed means and requires no iterative refinement. This simplicity makes TMR computationally attractive and effective. However, as discussed in Section 3.1.2, under strong heterogeneity this compression may collapse distinct local distributions into identical representatives, limiting its ability to capture local score geometry. Moreover, as a non-iterative construction, TMR does not adapt as the global model evolves.

SMR extends representative construction to an iterative framework by enforcing score-matching conditions that preserve response coherence. While this retains fidelity to local response ranges, SMR requires numerically solving for $\tilde{\eta}_k$, and the resulting solution may be non-unique. Moreover, the induced $\tilde{\mathbf{x}}_k$ can lie outside the bulk of local covariates, reducing representativeness and leading to slow or unstable convergence under low homogeneity, as demonstrated in Section 6.2.

Anchored-SMR addresses these limitations by introducing an anchoring mechanism that yields a unique, closed-form solution at each node. By stabilizing the intercept and constraining $\tilde{\mathbf{x}}_k$ within the local covariate support, Anchored-SMR preserves

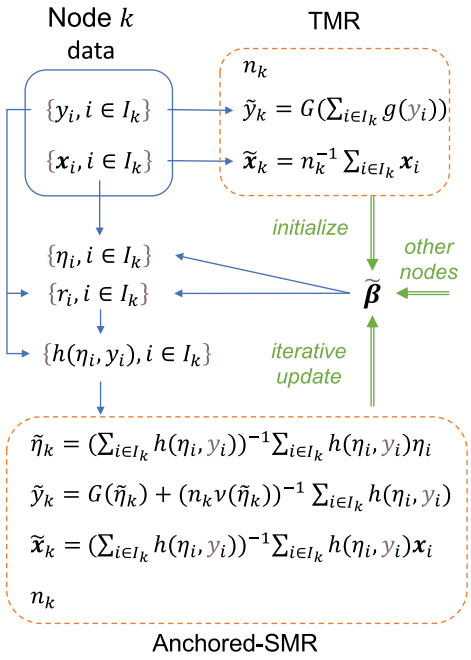


Figure 2. A flowchart for Node k showing how TMR and Anchored-SMR representatives are constructed from local data and current model information, with iterative updates in Anchored-SMR.

representativeness and achieves faster, more stable convergence in heterogeneous settings. The tradeoff is that the constructed $\tilde{y}_k$ may occasionally fall outside the empirical response range, particularly when residuals are highly unbalanced. This behavior is mitigated through subpartitioning, which decomposes local data into more homogeneous subgroups, ensuring residual alignment and improving interpretability.

From a computational perspective, TMR incurs a single pass over local data with complexity $O(n_k p)$ per node. SMR and Anchored-SMR are iterative, with per-iteration cost $O(n_k p)$; however, Anchored-SMR avoids numerical root-finding and typically converges in substantially fewer iterations with lower per-iteration overhead. As a result, Anchored-SMR offers a favorable balance between computational efficiency, stability, and adaptability, positioning it as a robust extension of TMR for asynchronous and heterogeneous distributed learning.

In RepL, each node transmits a fixed-size representative consisting of a pseudo covariate vector $\tilde{\mathbf{x}}_k \in \mathbb{R}^p$ and a scalar response $\tilde{y}_k$, resulting in $O(p)$ communication per update, independent of local sample size n_k. In contrast, gradient-based federated methods require transmitting full model updates or gradients of dimension $O(p)$ at every communication round, often across many rounds. Moreover, RepL supports asynchronous and event-driven communication, allowing nodes to transmit representatives only when local summaries change substantially, further reducing communication frequency. As a result, the total communication cost of RepL scales linearly with the number of representatives exchanged, rather than the number of local optimization steps.

The theoretical results in this section rely on standard regularity conditions commonly assumed in generalized linear models and convex optimization, including smoothness and strict concavity of the objective, which ensure identifiability and estimator stability. The consistency and likelihood preservation

guarantees for TMR and Anchored-SMR depend on controlling within-node heterogeneity through the covariate deviation Δ_k and residual spread R_k, both of which can be reduced in practice via subpartitioning or finer local grouping. The convergence results further assume bounded representative gradient approximation error ε_k, which naturally decreases as representatives more accurately capture local score information (Figure 3).

4. Generalizations and Extensions

This section introduces generalizations of the RepL framework across diverse settings, focusing on subpartitioning strategies for improved local fidelity, extensions to smooth and non-smooth objective functions, and incorporation of regularization terms in model fitting.

4.1. Subpartitioning for Local Fidelity

Subpartitioning improves local fidelity by controlling within-node heterogeneity in both covariates and model fit. As shown in Theorem 3.1, the error of TMR decreases with smaller covariate deviation Δ and transformed-response variation δ, motivating finer groupings with clustered covariates and stable response behavior.

Similarly, Theorem 3.2 for Anchored-SMR shows that the log-likelihood discrepancy between the full and representative datasets is bounded by $\sum_k n_k R_k$ and $\sum_k n_k \Delta_k R_k$, where R_k reflects residual variation and Δ_k denotes within-node covariate deviation. These bounds suggest that to improve representative fidelity, the focus is not on the size of individual Δ_k or R_k, but on minimizing their aggregated contributions across all nodes. One should iteratively subpartition nodes exhibiting high residual or covariate heterogeneity, thereby reducing the global sums $\sum_k n_k R_k$ and $\sum_k n_k \Delta_k R_k$.

For GLMs with constant variance function $v(\eta)$, partitioning by residual sign r_i helps control the correction term in the response representative $\tilde{y}_k = G(\tilde{\eta}_k) + n_k^{-1} \sum_{i \in I_k} r_i$ keeping $\tilde{y}_k$ near the observed range $[\min y_i, \max y_i]$. Moreover, consistent residual signs across a node ensure that the weights $h(\eta_i, y_i)$ in (7) are coherent, preserving the validity of the covariate representative $\tilde{\mathbf{x}}_k$.

If $\tilde{y}_k$ falls outside the feasible range or exhibits instability, further partitioning can be performed based on transformed responses (e.g., $\log(y_i)$ for Poisson models) or the linear predictors η_i. These additional splits further reduce both δ_k and Δ_k within each subpartition, improving representativeness and stability. For binary responses, where transformations are ill-defined, partitioning by class label naturally ensures coherent representatives.

These subpartitioning strategies enhance approximation quality and stability, as evidenced in Sections 6.1 and 6.7.

4.2. Generalization to Smooth Loss Functions

Consider learning problems where the loss function takes a general smooth form:

$$l(\boldsymbol{\beta}; \mathbf{y}, \mathbf{X}) = \sum_{i=1}^{n} F(y_i, \psi(\eta_i)),$$

where $F(y, \hat{y})$ quantifies the discrepancy between responses and predictions, and $\psi(\eta)$ maps the linear predictor $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$ to the predicted values, analogous to inverse link function in GLMs.

Both TMR and Anchored-SMR can be extended to this setting. For TMR, the response representative is constructed by applying $\psi(\cdot)$ to the average of transformed responses,

$$\tilde{y}_k = \psi(n_k^{-1} \sum_{i \in I_k} \psi^{-1}(y_i)), \quad \tilde{x}_{kj} = n_k^{-1} \sum_{i \in I_k} x_{ij}, \quad j = 1, \ldots, p.$$

For Anchored-SMR, the representative is constructed by matching local gradients. The representative values are computed as,

$$\tilde{\eta}_k = \frac{\sum_{i \in I_k} F_b(y_i, \psi(\eta_i)) \psi'(\eta_i) \eta_i}{\sum_{i \in I_k} F_b(y_i, \psi(\eta_i)) \psi'(\eta_i)}, \quad \tilde{\mathbf{x}}_k = \frac{\sum_{i \in I_k} F_b(y_i, \psi(\eta_i)) \psi'(\eta_i) \mathbf{x}_i}{\sum_{i \in I_k} F_b(y_i, \psi(\eta_i)) \psi'(\eta_i)},$$

where $F_b(a, b) = \partial F(a, b)/\partial b$, and $\psi'(\cdot)$ is the derivative of $\psi(\cdot)$.

These equations generalize the anchored score-matching approach to smooth losses, enabling representative construction for a broad class of objectives in statistical and machine learning models, including GLMs, Huber regression, and other smooth single-output losses. When $F_b(a, b)$ is monotonic in a, the solution for $\tilde{y}_k = \psi(\tilde{\eta}_k)$ is unique. Subpartitioning and regularization strategies developed for GLMs can be adapted to this setting to improve stability and accuracy.

However, this formulation does not accommodate all smooth objectives. In particular, models with vector-valued outputs (e.g., multinomial logistic regression, multivariate regression), or with nonlinear prediction mappings $f(\mathbf{x}; \boldsymbol{\theta})$, as in neural networks, fall outside the scope of this framework. These cases require structural modifications beyond the scalar $\psi(\eta)$ form assumed here.

4.3. Regularization Terms

In many applications, regularization is used to induce sparsity or stabilize estimation, as in the Lasso penalty $r(\boldsymbol{\beta}; \lambda) = \lambda \sum_j \beta_j$. However, representative construction relies on matching gradient structure or moment conditions, which involve linear predictors η_i and not the individual coefficients β_j. As a result, the regularization term cancels out in score-matching equations and does not affect representative generation.

Therefore, regularization is applied only during model fitting. After representatives are constructed, the global loss function (S.1), including regularization, is minimized to obtain the final estimate. Non-differentiable terms such as Lasso are handled using standard optimization tools in the final model stage.

Unlike RASMR (Zheng, Li, and Yang 2024), which uses cross-validation in each round, Anchored-SMR with regularization proceeds by incrementally updating model coefficients while holding λ fixed. The tuning parameter λ is selected via cross-validation conducted directly on TMR representatives at the central server. Since each representative carries a sample size weight n_k, this cross-validation is implemented by proportionally partitioning each representative's weight (e.g., 5-fold CV by splitting each weight into 5 shares). This weighted approach ensures that both the loss and penalty are properly scaled relative to the global objective. The resulting λ is then consistently

applied throughout the Anchored-SMR fitting stage without the need for rescaling or further tuning.

This approach proves effective in variable selection, as shown in Section 6.6, where MR, Anchored-SMR, and FULL are compared under varying noise levels for logistic regression models. The full Anchored-SMR procedure with regularization terms is detailed in Supplementary Algorithm S2.

5. Asynchronous and Decentralized Learning

Beyond centralized systems, real-world deployments often involve delays and restricted communication. We show how representative-based learning mitigates instability under asynchrony and enables scalable coordination over peer-to-peer networks.

5.1. Asynchronous Updates

In distributed learning, synchronization between nodes is crucial for maintaining consistency in model updates. When nodes are nonsynchronized, the gradient directions in SGD can misalign with the current coefficient values, causing failure. Unlike pure gradient-based methods, Anchored-SMR carries richer information beyond gradients, but nonsynchronization can still impact it, especially when information is heavily delayed.

To understand this effect, a first-order expansion shows that the error incurred by using a stale representative is proportional to the coefficient drift $|\Delta\boldsymbol{\beta}|$ and the local gradient magnitude, motivating the following theoretical analysis.

Theorem 5.1. Suppose the global objective $f(\boldsymbol{\beta})$ is μ-strongly convex with L-Lipschitz continuous gradient. Each node k provides a representative $\tilde{D}_k^{(t-\tau_k(t))}$ constructed with delayed model $\boldsymbol{\beta}^{(t-\tau_k(t))}$, where $\tau_k(t) \leq \tau_{\max}$. Let ε_k denote the representative gradient approximation error at node k, and assume that per-node gradients under representative data are uniformly bounded by $\|\nabla l_k(\boldsymbol{\beta}; \tilde{D}_k)\| \leq M$.

Then, for step size $\eta \leq \min\left\{\frac{\mu}{2L^2}, \frac{1}{8L(\tau_{\max}+1)}\right\}$, the iterates satisfy

$$\mathbb{E}\left[\|\boldsymbol{\beta}^{(T)} - \boldsymbol{\beta}^*\|^2\right] \leq (1-\eta\mu)^T \|\boldsymbol{\beta}^{(0)} - \boldsymbol{\beta}^*\|^2 + O\left(\frac{\eta}{\mu}\left(L^2\tau_{\max}^2 M^2 + \varepsilon^2\right)\right),$$

where $\boldsymbol{\beta}^*$ is the global minimizer and $\varepsilon = \sum_{k=1}^K \frac{n_k}{n}\varepsilon_k$.

Anchored-SMR converges linearly up to a residual floor induced by gradient delay ($\tau_{\max}$) and representative approximation error (ε). See Supplementary Section S.9 for a complete proof.

The asynchronous framework also naturally accommodates partial participation, where only a subset of nodes transmit representatives at a given iteration. In this case, nonparticipating nodes can be viewed as having unbounded or random delays, and the convergence analysis continues to hold under the same bounded-delay or bounded-staleness assumptions on participating nodes. As a result, Anchored-SMR remains stable even when node participation is intermittent or uneven, which is common in large-scale or resource-constrained distributed systems. Theorem 5.1 also applies when only a subset of nodes participates at each iteration, provided that each node contributes infinitely often and delay bounds hold for participating updates.

To empirically validate the theoretical robustness of Anchored-SMR under delay, we simulate asynchronous updates where a fixed proportion of nodes transmit representatives constructed from delayed models. Anchored-SMR consistently outperforms Federated Averaging, maintaining fast convergence even under severe delays (Figure 2(d)). This confirms its practical advantage in environments with irregular communication.

5.2. Decentralized Learning via Anchored-SMR

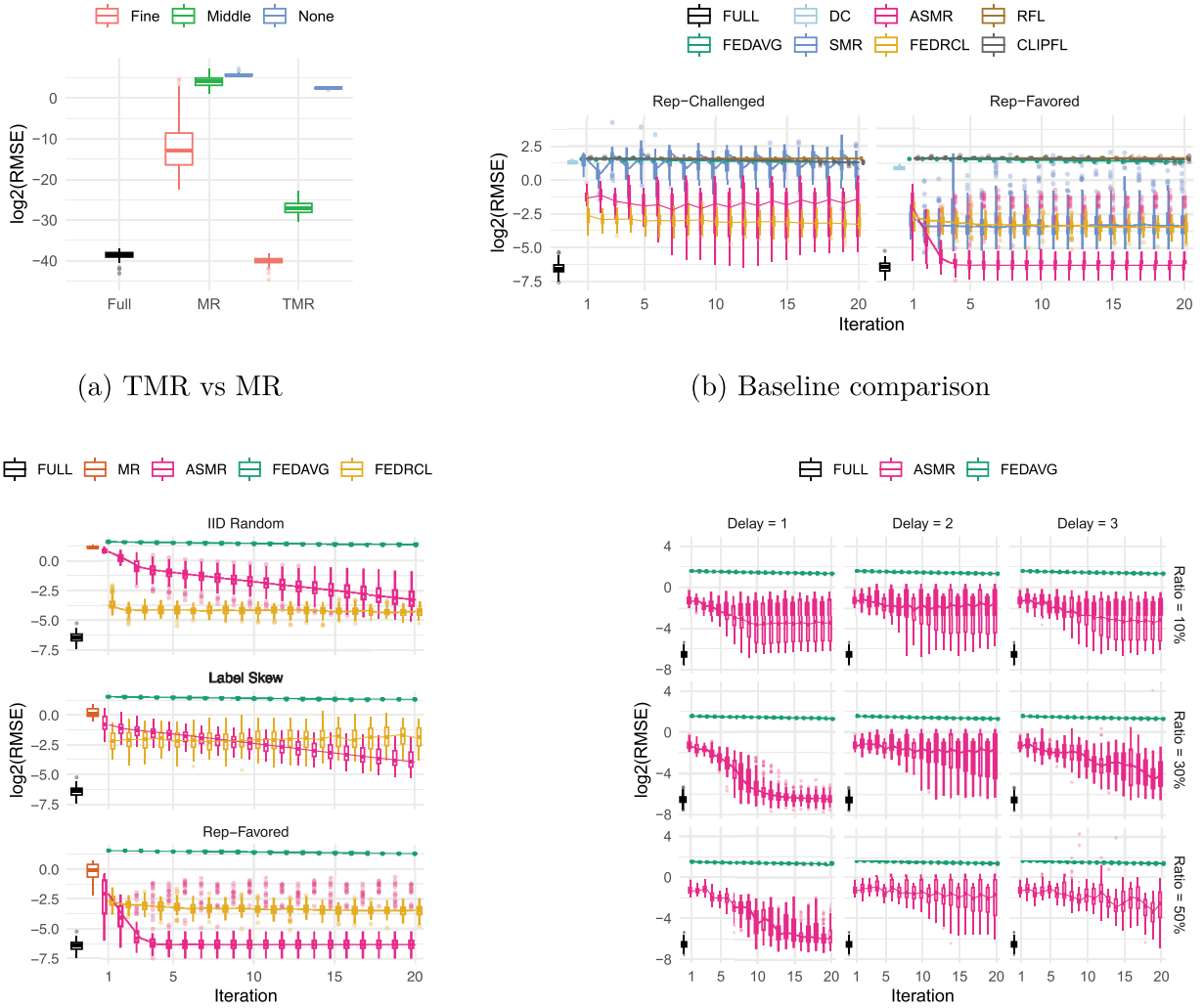
We extend Anchored-SMR to decentralized learning environments without a central server, where nodes operate over a dynamic peer-to-peer network. Two communication protocols are considered, depending on whether nodes are permitted to exchange model parameters.

In the first scenario, at each communication round t, each node k maintains its local model $\boldsymbol{\beta}_k^{(t)}$ and representative $\tilde{D}_k^{(t)}$, and transmits both to a time-varying neighborhood $\mathcal{N}_k^{(t)} \subset \{1,\ldots,K\} \setminus k$. Upon receiving $\{\boldsymbol{\beta}_\ell^{(t)}, \tilde{D}_\ell^{(t)}\}_{\ell\in\mathcal{N}_k^{(t)}}$, node k averages model coefficients to form an temporal $\check{\boldsymbol{\beta}}_k^{(t)}$, then updates its model using a mini-batch from local data $B_k^{(t)}$ and the set of received representatives. The updated model $\boldsymbol{\beta}_k^{(t+1)}$ is used to compute a new representative $\tilde{D}_k^{(t+1)}$, maintaining synchronization with local learning progress. This approach enhances convergence by combining structural summaries with limited peer model sharing under decentralized and heterogeneous conditions.

In the second scenario, model sharing is restricted: nodes transmit only their representatives to neighbors. Local updates to $\boldsymbol{\beta}_k^{(t)}$ are then performed using $B_k^{(t)}$ and $\{\tilde{D}_\ell^{(t)}\}_{\ell\in\mathcal{N}_k^{(t)}}$, without access to peer model parameters. This setting arises when privacy policies forbid model disclosure or when model sizes render transmission impractical. Even without model exchange, the structural information encoded in representatives supports implicit alignment and cross-node learning, enabling effective model training in constrained environments.

To account for asynchrony and communication delays, we introduce a time-decay scheme that discounts outdated representatives. Let $\tau_t(k,\ell) \leq t$ denote the most recent round when node k received information from node ℓ. The representative $\tilde{D}_\ell^{(\tau_t(k,\ell))}$ is weighted by a decay factor $w_{k,\ell}^{(t)} = \gamma^{t-\tau_t(k,\ell)}$ for some $\gamma \in (0,1)$, ensuring that recent updates are prioritized while still leveraging historical summaries. These weights are used in both representative aggregation and local model fitting, stabilizing convergence under delayed and uneven communication.

This decentralized algorithm, referred to as Anchored-SMR with Decentralized Communication (ASMR-DC), provides a scalable and communication-efficient framework that supports heterogeneous models, partial connectivity, and time-varying topology. See Algorithm S3 in the Supplementary Materials for details.



(a) TMR vs MR

(b) Baseline comparison

(c) Partitioning strategies

(d) Asynchronous updates

Figure 3. Comparison of various methods and scenarios. (a) TMR vs MR: $\log_2$ (RMSE) comparison for FULL, MR, and TMR across partitioning strategies, where TMR remains stable while MR deteriorates under increasing within-node heterogeneity. (b) Baseline comparison over 20 iterations: Anchored-SMR steadily reduces error, narrows the gap with FedRCL under IID splitting, surpasses it under label-skew, and performs best in the Rep-Favored regime, while FedAvg converges more slowly. (c) Robustness of Anchored-SMR across IID, label-skew, and conditional-shift partitioning, showing stable convergence. (d) $\log_2$ (RMSE) trajectories under asynchronous updates with varying delay and participation levels, where Anchored-SMR maintains higher accuracy despite delays.

6. Simulation Studies

We evaluate representative learning methods under logistic and Poisson regression models. Each method is assessed using RMSE between estimated and true coefficients, averaged over 100 replicates.

Covariates $\mathbf{x}_i \in \mathbb{R}^{10}$ are sampled from diverse distributions (e.g., Gaussian, t, Beta), with true coefficients $\boldsymbol{\beta}$ normalized to $\|\boldsymbol{\beta}\|_2 = 3$. Unless stated otherwise, we use $K = 1000$ nodes, each with $n = 1000$ samples, with responses drawn from logistic or Poisson models.

We consider two heterogeneity regimes: Rep-Favored (low within-node variability, high between-node diversity) and Rep-Challenged (high within-node variability, low between-node diversity). Within each regime, subpartitioning strategies **None**, **Middle**, and **Fine** control intra-node representative granularity, with Anchored-SMR using residual-sign splits and SMR using residual-plus-η splits. These settings disentangle robustness to inter-node distribution shift and intra-node compression. Additional partitioning experiments are reported below, with full details in Supplementary Section S.3.

6.1. TMR vs MR

Figure 2(a) shows that under Poisson regression in the Rep-Challenged scenario, TMR significantly outperforms MR, especially under coarse or no subpartitioning. While MR performs well only under Fine subpartitioning, TMR remains stable across all three within-node partitioning levels (None, Middle, and Fine). This confirms TMR's robustness to within-node dispersion and to the choice of within-node partitioning granularity, as well as its correction of nonlinear link effects. The advantage is most pronounced when local responses vary significantly, consistent with theoretical results.

6.2. Comparison with Baseline Methods

We compare Anchored-SMR (ASMR) with SMR, FedAvg, DC-GLM, and recent federated learning baselines including FedRCL, Robust Federated Learning (RFL), and CLIP-guided Federated Learning (CLIPFL) under logistic regression in both Rep-Challenged and Rep-Favored regimes. Figure 2(b) reports $\log_2$ (RMSE) trajectories over 20 iterations.

Across both regimes, FedAvg, DC-GLM, RFL, and CLIPFL exhibit relatively slow error reduction and remain substantially above the FULL-data benchmark. Their performance deteriorates further under strong within-node variability, indicating sensitivity to gradient mismatch and local instability. Under the Rep-Challenged regime, FedRCL achieves lower RMSE than ASMR, benefiting from contrastive alignment that partially mitigates within-node variability. However, in the Rep-Favored regime characterized by pronounced between-node heterogeneity, ASMR consistently outperforms FedRCL and converges closer to the FULL benchmark. This contrast highlights the advantage of representative-based communication when heterogeneity arises from structural distribution mismatch rather than feature misalignment alone. Table S1 (Supplementary) shows a representative replicate under extreme variability, where SMR exhibits instability and large RMSE inflation, whereas ASMR remains stable. Overall, ASMR demonstrates competitive or superior accuracy across heterogeneous regimes while maintaining stable convergence behavior.

6.3. One-Step vs Full Optimization

Using logistic regression, we compare one-step Anchored-SMR updates versus full optimization at each iteration in the Rep-Challenged scenario. As shown in Supplementary Figure S1a, the one-step strategy exhibits stable RMSE across all iterations, while full optimization suffers from spikes due to overreacting to transient gradients. In some replicates, full optimization diverges entirely (see Table S2, Supplement). These findings support our use of the incremental update rule.

6.4. Various n and K

We study the effect of node size $n \in 100, 1000, 10000$ and number of nodes $K \in 100, 1000$ under logistic regression (Rep-Favored). Supplementary Figure S1b shows that both MR and ASMR benefit from larger n and K, with ASMR converging closer to FULL baseline under large n. The trend confirms statistical consistency under increasing data.

6.5. Robustness to Data Partitioning Strategies

To evaluate sensitivity to node construction rules, we compare MR, Anchored-SMR, FedAvg, and FedRCL under three partitioning strategies: (i) IID random split, (ii) label-skew split, and (iii) feature/conditional-shift split (the Rep-Favored regime used above). For (iii), we reuse the existing Rep-Favored results to avoid redundant simulations. As shown in Figure 2(c), Anchored-SMR exhibits stable convergence across all partitioning strategies, with performance increasingly improving over iterations. Under IID random splitting, Anchored-SMR initially trails FedRCL but progressively reduces the gap as iterations proceed. Under label-skew partitioning, where class proportions vary substantially across nodes, Anchored-SMR surpasses FedRCL after several iterations and maintains lower RMSE thereafter. In the Rep-Favored regime, Anchored-SMR consistently outperforms FedRCL and approaches full-data performance, demonstrating robustness to structured feature and conditional shifts. In contrast, FedAvg converges more slowly across all partitioning strategies and shows comparatively limited improvement over iterations. Overall, the advantage of representative-based communication becomes more pronounced as partition-induced heterogeneity increases.

6.6. Variable Selection

Under high-dimensional logistic regression ($p = 10$ signal, $q \in 10, 20, 50, 100$ noise), we evaluate variable selection via ℓ_1-regularization. MR consistently achieves the lowest false positives and negatives, particularly in the Fine regime (Supplementary Figure S1c). Although ASMR improves estimation, MR provides cleaner variable selection even comparable to FULL in sparse models, likely due to less sensitivity to gradient magnitude.

6.7. Poisson Regression

ASMR and TMR are evaluated on Poisson regression under Rep-Challenged data, with Fine and Middle subpartitioning strategies. As shown in Supplementary Figure S1d, Fine outperforms Middle and closely aligns with the FULL estimate.

6.8. Asynchronous Updates

Figure 2(d) illustrates the performance of ASMR and FedAvg under asynchronous communication, where 10–50% of nodes transmit outdated representatives (delays of 1–3 iterations). Under Rep-Challenged data, ASMR maintains steady improvement and approaches the FULL baseline, even under substantial delay. FedAvg stagnates or degrades with delay. In Rep-Favored settings (Figure S1e, Supplement), ASMR recovers rapidly after initial drift, except in a few rare cases, demonstrating high resilience.

7. Airline On-Time Performance Data

We apply the Anchored-SMR method to the Airline On-Time Performance dataset, which tracks US domestic flight arrivals from October 1987 to December 2019, provided by the Bureau of Transportation Statistics, following the setup in Li and Yang (2022). After cleaning, the dataset consists of over 188 million valid records. We model the binary response `ArrDel15` (arrival delayed by 15 minutes or more) using a logistic regression model with covariates `QUARTER` (season), `DayOfWeek` (day of the week), `DepTimeBlk` (departure time block), and `DISTANCE` (scaled by 1000 for comparability).

We generate the response using a predefined coefficient vector $\boldsymbol{\beta}_{\text{true}}$ (Li and Yang 2022) and partition the dataset naturally by month. Each file is further split by `DayOfWeek`, `DepTime-`

Table 1. RMSE comparison for airline on-time performance data (3rd iteration).

Method	Mean	Std. Dev.
SMR	0.428	0.045
Anchored-SMR	0.048	0.002

`Blk`, and the sign of h_i (Anchored-SMR), or by the sign of the linear predictor and response (SMR). To account for variability, we repeat the response generation 10 times, producing replicates of the experiments to compute standard deviations.

Since full data estimates are computationally infeasible due to the dataset size, we compare Anchored-SMR and SMR using the Root Mean Squared Error (RMSE) based on the generated response. Both methods are evaluated at the 3rd iteration of their respective fitting processes. As shown in Table 1, Anchored-SMR consistently achieves a lower mean RMSE compared to SMR, indicating its better performance for heterogeneous data distributions.

For more details, see Section S.10 in the Supplementary Material.

8. Conclusion and Future Directions

This work establishes RepL as a principled framework for distributed learning, where nodes transmit structured pseudo data instead of model updates. We develop two constructions, TMR and Anchored-SMR, that enable robust and interpretable learning under heterogeneity and asynchrony.

Anchored-SMR demonstrates strong empirical performance and theoretical guarantees under smooth convex losses, bounded delays, and gradient approximation errors. Simulation results confirm that it consistently outperforms FedAvg across synchronous and asynchronous regimes, even under high variability or communication delays.

RepL is well suited for large-scale and resource-constrained systems, as representative construction and updates are performed entirely locally with no synchronization requirement. Representatives can be generated in a streaming fashion and stored or transmitted asynchronously, making the framework compatible with parameter-server, peer-to-peer, or edge-computing architectures. Since representatives are model-agnostic pseudo data, they can be reused across different downstream models or tasks without recomputation, reducing system overhead. In practice, subpartitioning can be used to balance representational fidelity and communication cost, enabling adaptive deployment in large and heterogeneous networks.

Despite these advantages, the current RepL framework has several limitations. First, existing representative constructions are tailored to smooth, differentiable loss functions, and their theoretical guarantees rely on convexity or local smoothness assumptions. As a result, direct application to non-smooth or non-convex objectives may lead to unstable or poorly informative representatives. Second, while bounded delay and partial participation are accommodated in the analysis, extreme asynchrony, adversarial behavior, or highly unbalanced participation patterns remain challenging and are not fully characterized within the current framework. Third, representative construction is currently decoupled across nodes, which may limit adaptivity in fully decentralized or rapidly evolving environments. In addition, the current analysis focuses on single-task learning with continuous covariates, and does not explicitly address multi-task settings or structured categorical features beyond standard encodings.

To further advance the RepL framework, several methodological challenges remain. Extending RepL to non-smooth objectives such as hinge loss or quantile regression will require new representative designs beyond score-matching, potentially involving subgradient-based or distributional summaries. Enhancing robustness to unbounded delay, adversarial behavior, and highly intermittent participation calls for additional stabilization mechanisms and diagnostic tools. Moreover, scaling RepL to decentralized and online systems motivates dynamic and adaptive representative construction strategies that evolve with data streams and network conditions. While Section S.11 introduces representative-based heterogeneity metrics, further work is needed to integrate these tools into adaptive learning and system-level decision-making pipelines. Extending RepL to multi-task learning scenarios, where nodes solve related but non-identical objectives, as well as to mixed-type data with richer categorical structure beyond standard encodings, represents a natural and important direction for future work.

Taken together, our findings suggest that RepL offers a promising and versatile foundation for communication-efficient, privacy-preserving, and interpretable distributed machine learning.

Supplemental Materials

Supplementary Text: Contains background on GLM score matching (Section S.1), full algorithmic procedures (Section S.2), simulation configurations (Section S.3), additional figures and tables (Sections S.4–S.5), transformed means for common GLMs (Section S.6), TMR limitations in binary data (Section S.7), Anchored-SMR examples (Section S.8), theoretical proofs (Section S.9), and details on the airline delay application (Section S.10). Section S.11 presents RepL extensions to model heterogeneity. (PDF)

RepL R Code: Reproducible R implementation of MR, TMR, SMR, and Anchored-SMR (managed via `renv`), including scripts to regenerate all simulation results and figures. (zip)

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Data Availability Statement

The Airline On-Time Performance data used in Section 7 are publicly available from the Bureau of Transportation Statistics: https://www.transtats.bts.gov/DataIndex.asp. Simulation data are too large to upload but are available from the authors upon reasonable request or after the review process.

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